

Introduction to Economics

Class 6 — Equilibrium, shifts and prices as signals

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Slide 1 Title

Preview version. The full slides and handout are posted after class.

Class 6 puts Monday's two curves on one picture. The first hour: where they cross, what happens when the price sits above or below the crossing, and how the price does the work that no planner does. Then what happens to the crossing when a curve moves, in three steps that return in every problem of this kind from now on. The thread through the class is a real price: olive oil at origin, 3.60 euros a kilo in July 2022 and 9.20 in January 2024.

Plan for today

Monday, one more look
Olive oil: what happened?
Where the two curves cross
Above and below the crossing
Prices as signals
When a curve moves: three steps
Break
Idea meetings today, during class
Supply moves; both curves move
Nine cases, and a teaser
Before Thursday 15 October

Slide 2 Plan for today

The class at a glance. First a last look at Monday's rule; then a real price that more than doubled, and the question of which curve moved. The crossing of the two curves, what happens above and below it, the price as a signal, and the three steps for a curve that moves. After the break the idea meetings, which happen today during class; then supply moves, both curves at once, nine cases and a teaser for next week.

Two halves of fifty minutes with a ten-minute break between them.

Monday, one more look: milk gets dearer — which curve moves?

The neighbourhood's market for *café amb llet*. Milk goes up in price.

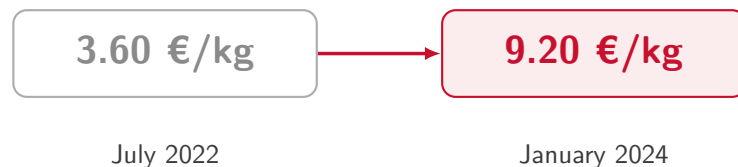
- A) Demand: a dearer cup, fewer buyers
- B) Supply: an input got dearer
- C) Both curves
- D) Neither: nobody changed the price of a cup

Slide 3 Monday, one more look: milk gets dearer, which curve moves?

Monday's example once more: milk gets dearer in the neighbourhood's market for *café amb llet*. Before class, decide which curve moves, if any, and say in one sentence what changed for the side you chose and what did not change for the other. Use Monday's test: what changed, and for whom?

Revealed in class.

Olive oil at origin: 3.60 euros a kilo, then 9.20



What happened — and which curve?

Poolred, the Spanish olive-oil origin price system: weekly prices paid to mills for extra virgin oil.

Slide 4 Olive oil at origin: 3.60 euros a kilo, then 9.20

The hook. The price at origin is what the mills get, before bottling and shops. Extra virgin olive oil cost 3.60 euros a kilo in July 2022 and 9.20 in January 2024, the highest ever recorded: two and a half times the price in eighteen months. What happened, and, in Monday's words, which curve? The next slide answers.

What happened

Revealed in class

Before class: write down your guess — was it the buyers' side or the sellers' side, and what was the event?

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Slide 5 What happened

Revealed in class.

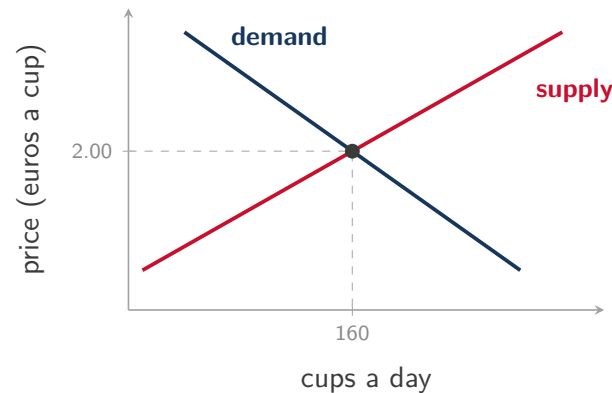
Beforehand: write down your guess. Was it the buyers' side or the sellers' side, and what was the event? Then ask what the price would do if the event were reversed.

Where the two curves cross: 160 cups a day at 2.00 euros

price	wanted	offered
1.00	240	60
1.50	200	110
2.00	160	160
2.50	120	210
3.00	80	260

the neighbourhood, cups a day

Equilibrium price: the one price at which wanted = offered. No queue, nothing left over.



Slide 6 Where the two curves cross: 160 cups a day at 2.00 euros

Monday's last picture, now with numbers: the whole neighbourhood's buyers (navy) and sellers (red) of café amb llet, in cups a day. The table lists what buyers want and what sellers offer at each price, and only one row agrees: at 2.00, 160 cups wanted and 160 offered. That price is the equilibrium price, or market price, and 160 cups the equilibrium quantity.

At 2.00 every buyer who wants a cup at that price gets one and every seller who wants to sell at that price sells: no queue, nothing left over. The market clears. The word is borrowed from physics, a state of rest with no force pushing; the next two slides are the forces.

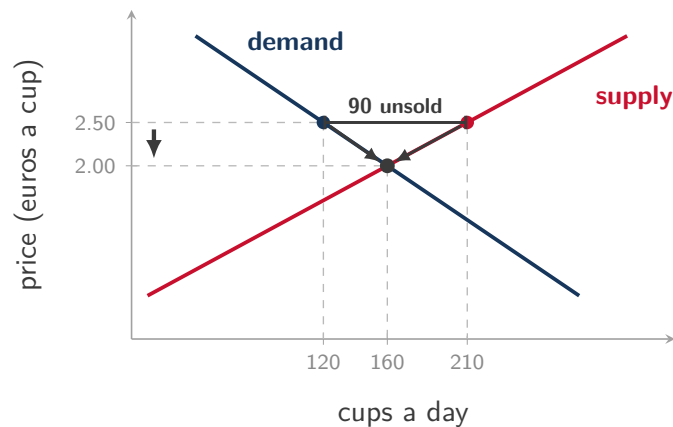
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At 2.50: offered 210, wanted 120 — a surplus of 90



Surplus: unsold cups. Sellers cut the price; down both curves to 2.00.

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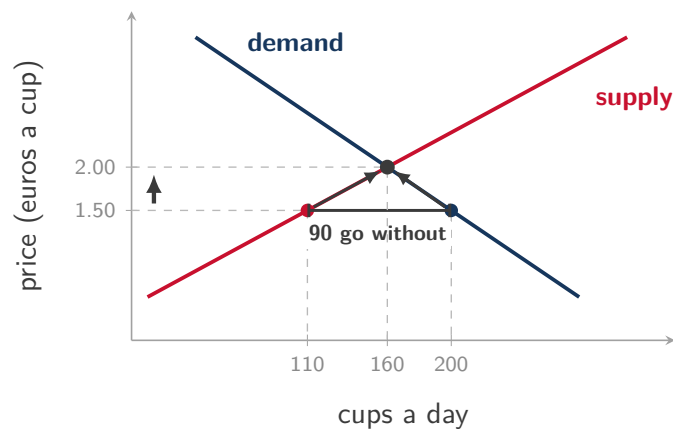
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Slide 7 At 2.50: offered 210, wanted 120, a surplus of 90

Suppose the bars try 2.50. Buyers want 120 cups, sellers offer 210: 90 cups a day unsold. That is a surplus, or excess supply: milk poured away, staff standing around. A seller with unsold cups cuts the price; as it falls, buyers come back down their curve and sellers make fewer along theirs, and the gap closes at 2.00.

The two small arrows run along the curves. Nothing shifted: the price moved, and both sides slid along their own curve.

At 1.50: wanted 200, offered 110 — a shortage of 90



Shortage: a queue. Sellers raise the price; up both curves to 2.00.

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Slide 8 At 1.50: wanted 200, offered 110, a shortage of 90

The mirror image at 1.50. Buyers want 200 cups, sellers offer 110: a shortage, or excess demand, of 90: a queue, people turned away, the sold-out sign. A seller with a queue raises the price without losing sales; some buyers drop out up their curve, sellers make more up theirs, and the market is back at 2.00.

The reading calls this the law of supply and demand: the price adjusts until the quantity supplied and the quantity demanded are equal. Comparing one resting point with another is what the book calls comparative statics, the method of the second hour.

In pairs, three minutes: bikinis at the bars around the faculty

Where does the market settle? What happens at 5.50, and at 4.00?

price (€)	wanted	offered
4.00	300	80
4.50	250	140
5.00	200	200
5.50	150	260
6.00	100	320

bikinis a day

- 1 The crossing:
- 2 At 5.50:
- 3 At 4.00:

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Slide 9 In pairs: bikinis at the bars around the faculty

Exercise 1: the bikini, the toastie sold at the bars around the faculty, in bikinis a day. Before class, find the row where wanted equals offered; then, for 5.50 and for 4.00, work out how many are wanted, how many offered, the size of the gap and which way the price moves. Say also how many bikinis actually change hands at each of those two prices.

Revealed in class.

Nobody counts the cups. The price does.

The price rises

buyers: fewer cups, or tea
sellers: more cups, longer hours

The price falls

buyers: more cups
sellers: fewer cups, shorter hours

Prices as signals: one number tells buyers what to give up and sellers what it pays to make.
160 cups a day, planned by nobody.

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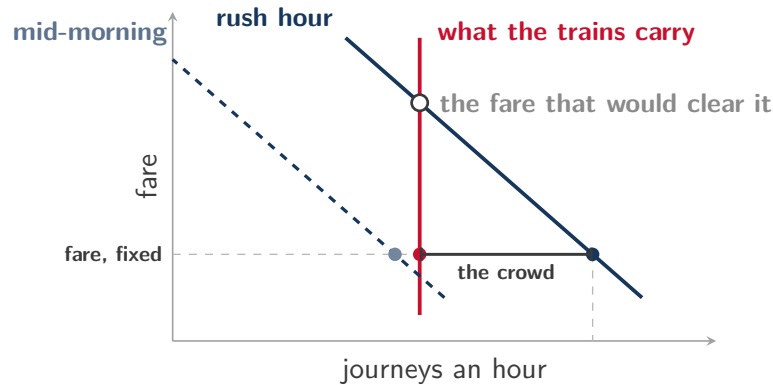
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Slide 10 Nobody counts the cups. The price does.

No office decided on 160 cups a day. The price did it, and it does two jobs at once. To buyers it says what a cup costs in other things given up: at 2.50 some decide a tea or a walk is worth more. To sellers it says what it pays to make: at 2.50, a second barista and longer hours. A rising price tells sellers to make more and buyers to give something up; a falling price says the reverse.

One number, seen by everyone, read by everyone, chosen by no one: a decentralised market. Whoever is willing and able to pay the price gets the good; whether that is fair is a normative question, and the model only says what happens.

When the price cannot move, the crowd does the rationing



A fare set by decree: the shortage is the full platform, not a price. Price controls: Class 9.

Slide 11 When the price cannot move, the crowd does the rationing

Monday's metro, where the price cannot move: the fare is set by the transport authority. The red line is what the trains can carry in an hour; the navy curve is the journeys people want at each fare at rush hour, the dashed one mid-morning. At the fixed fare, rush-hour demand runs past the capacity: the gap is the crowd on the platform, a shortage that cannot become a higher price, so it becomes waiting. The hollow circle is the fare that would clear the platform; nobody proposes it. When the price is fixed, something else does the rationing: queues, crowding, first come first served. Price controls: Class 9.

When a curve moves

Three steps, and what the price does

Slide 12 When a curve moves

The second half. Monday was the curves; now what happens to the crossing when one of them moves.

A change in the market: three steps, always the same

1

Which curve?

buyers' side: demand
sellers' side: supply
sometimes both

2

Which way?

right: more, left: less
at every price

3

The new crossing

draw it; read off
the price, the quantity

Step 1 is Monday's rule: anything but the good's own price moves a curve. The price is step 3.

Slide 13 A change in the market: three steps, always the same

The method for every question of this kind from now on. Step 1, which curve: the side the event touches, the buyers' (demand) or the sellers' (supply), sometimes both. Step 2, which way: right if more is wanted or offered at every price, left if less. Step 3: draw the new crossing and read off the new price and quantity.

Step 1 is Monday's rule: anything but the good's own price moves a curve. Step 3 is where the price comes in: it is the result, never the event. The common mistake is to skip step 3 and guess. Draw it.

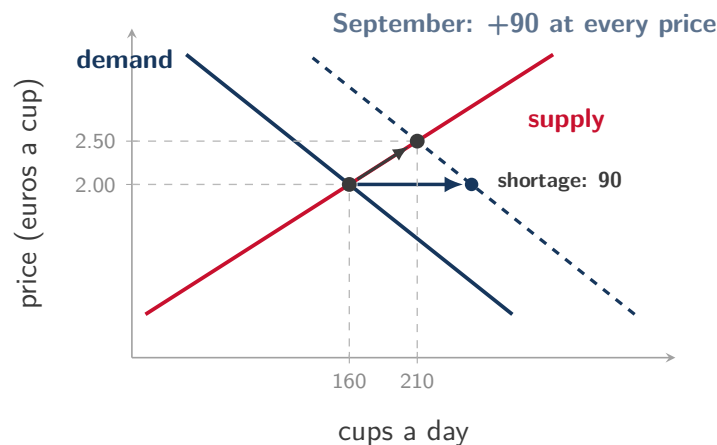
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Demand shifts right: a dearer cup, and more of them



Demand moved, supply did not: the market slid *up the supply curve* to 2.50 and 210 cups.

Slide 14 Demand shifts right: a dearer cup, and more of them

The coffee market at rest, 160 cups at 2.00, and an event: students are back in September, so there are more buyers. Step 1: demand. Step 2: right, the dashed curve, 90 more cups wanted at every price. At the old price buyers now want 250 and sellers offer 160, a shortage of 90, and slide 8 says what a shortage does: the price rises. Step 3: as it rises, sellers move up their supply curve and some of the new buyers drop out along the new demand curve; the new crossing is 2.50 and 210 cups.

The sentence to keep: demand moved, supply did not. The quantity supplied rose, along the same red line. Price and quantity both up.

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Slide 15 Break

Ten minutes. Sit with your project group afterwards: the idea meetings happen during class, and I call each group in turn.

Break

Ten minutes

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Idea meetings: today, during class

- Six minutes per group; I call you in class
- Bring your paragraph: we talk about the question, not the topic
- Topic fixed by Thursday 15 October



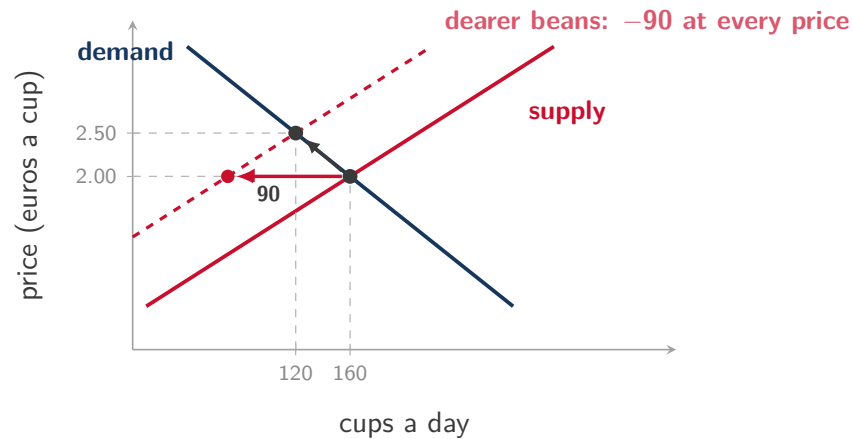
course page

Slide 16 Idea meetings: today, during class

The idea meetings are today, during class: six minutes per group, in the order on Monday's sign-up sheet; I call each group in turn, while the class works on an exercise in pairs or in a few spare minutes. Bring your paragraph in your head. The six minutes are for the question, whether it is specific, local and something you can observe or measure before the poster session, not for explaining the topic. A group whose paragraph has not arrived sends it or comes anyway.

The topic is fixed by Thursday 15 October; the poster session is on Thursday 5 November. Anyone still without a group comes to the meetings too.

Supply shifts left: a dearer cup, and fewer of them



Supply moved, demand did not: the market slid *up the demand curve* to 2.50 and 120 cups.

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In pairs: the coffee market — which curve, which way, and then?

- 1 Milk gets cheaper.
- 2 A heatwave: fewer people want a hot drink.
- 3 Students are back in September.
- 4 Baristas' wages rise.
- 5 A cup gets dearer.

Four minutes. Which curve, which way — and what happens to the price and the quantity?

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Slide 17 Supply shifts left: a dearer cup, and fewer of them

The same market at rest and a second event: coffee beans get dearer, an input, from Monday's list of what moves supply. Step 1: supply. Step 2: left, 90 fewer cups offered at every price. At the old price sellers offer 70 and buyers want 160: a shortage of 90 again, so the price rises. Step 3: buyers move up their demand curve and buy fewer cups; the new crossing is 2.50 and 120 cups. Demand did not move.

Compare with slide 14: both events raised the price to 2.50, but one raised the quantity and the other lowered it. Which curve moved shows in the quantity, not in the price.

Slide 18 In pairs: the coffee market, which curve, which way, and then?

Exercise 2: five events on the neighbourhood's coffee market. Before class, take the three steps for each one: which curve, which way, and what then happens to the price and to the number of cups sold. For each event ask whose side it touches, the buyers' or the sellers', and whether anything moved a curve at all.

Revealed in class.

Back to the olive oil

Revealed in class

Before class: take the three steps for the oil — which curve, which way, and what the new crossing says about the price and the kilos sold.

Slide 19 Back to the olive oil

Revealed in class.

Beforehand: take the three steps for the olive oil. Which curve did the event touch, which way did it move, and what does the new crossing say about the price and about the kilos sold? Then ask what happens when the event is reversed.

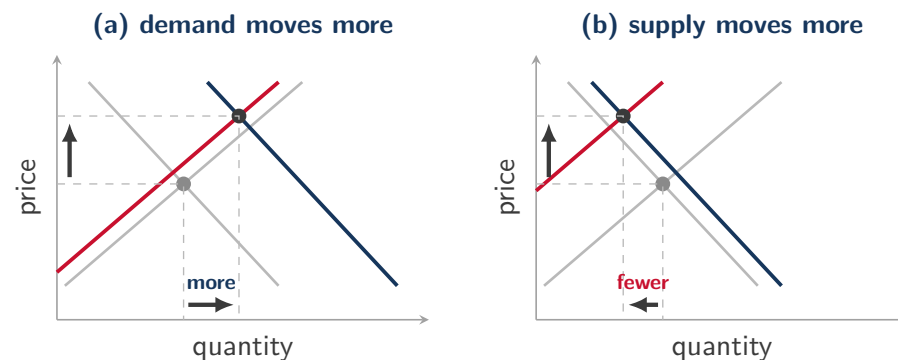
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Demand right, supply left: the price rises for sure; quantity depends



In grey, the curves before. Both shifts push the price up; they pull the quantity opposite ways.

Slide 20 Demand right, supply left: the price rises for sure; quantity depends

Both curves at once: demand right and supply left, slides 14 and 17 together, students back and dearer beans; the grey lines are the curves before. Both events push the price up, so the price rises for sure. The quantity is pulled both ways, up by the demand shift and down by the supply shift, and which wins depends on how far each curve moved. Panel (a), demand moved more: more cups. Panel (b), supply moved more: fewer.

So the honest answer for the quantity is that it depends on the sizes, which the model does not know and the data do. Writing that the price rises and the quantity could go either way is a full answer; writing that both rise is wrong.

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In pairs: 23 April, the day of the roses

Everyone wants a rose; florists fly them in. The model says:

- A) Dearer roses, and more of them
- B) More roses for sure; the price: it depends
- C) Dearer roses for sure; how many: it depends
- D) Nothing for sure

Slide 21 In pairs: 23 April, the day of the roses

Exercise 3: 23 April, when everyone wants a rose and florists fly them in from everywhere. Before class, take the three steps for the buyers' side and for the sellers' side separately, then put the two shifts together: what does the model promise for the number of roses, and what for the price? Compare with what you remember paying.

Revealed in class.

One curve moves: both known. Both move: one of them depends

	supply unchanged	supply right	supply left
demand unchanged	no change	$P \downarrow \cdot Q \uparrow$	$P \uparrow \cdot Q \downarrow$
demand right	$P \uparrow \cdot Q \uparrow$		
demand left	$P \downarrow \cdot Q \downarrow$		

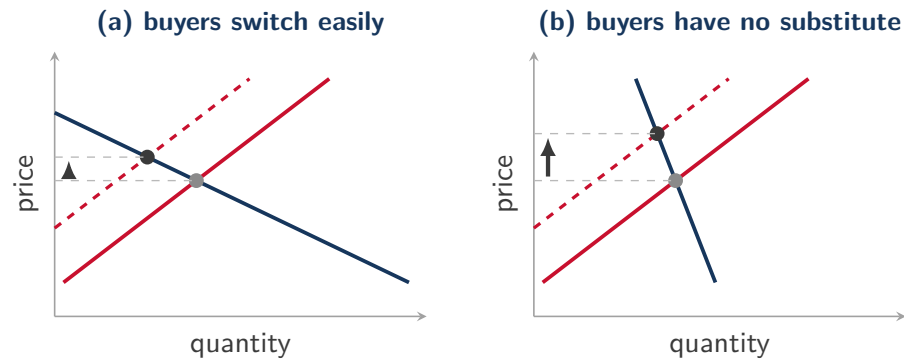
Four cells to fill in before class: draw each crossing twice, demand moving more, then supply moving more.

Slide 22 One curve moves: both known. Both move: one of them depends

The nine cases in one table: the row is what demand did, the column what supply did. When one curve moves, the first row and the first column, both answers are known. The four cells where both curves move are left for you: before class, draw each of them twice, once with demand moving more and once with supply moving more, and note which of price and quantity comes out the same both times and which does not.

Do not memorise the table. Draw the crossing and the table writes itself.

The same shift, two markets: how far does the price jump?



Which way is today's question. How much is Thursday 15 October's: **elasticity**.

Slide 23 The same shift, two markets: how far does the price jump?

The teaser: the same supply shift to the left in two markets. In (a) buyers switch easily and the demand curve is flat: the price barely moves and the quantity falls a lot. In (b) buyers have no substitute and the demand curve is steep: the price jumps and the quantity hardly falls. Before class, think of one good you would keep buying at almost any price and one you would drop at once, and say which panel each belongs to.

Today's question is which way. Next week's is how much: elasticity, the reading for Thursday 15 October.

Slide 24 In one line

The class in one line: the price moves until wanted equals offered; when a curve moves, find the new crossing. One curve moving: price and quantity both known. Both curves: one of the two depends on which moved more.

**The price moves until wanted equals offered.
A curve moves: find the new crossing.**

One curve: price and quantity both known.
Both curves: one of them depends on which moved more.

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Before Thursday 15 October

- No class Monday 12 October (Festa Nacional)
- Read Mankiw & Taylor, Ch. 3, pp. 53–66 — price elasticity of demand, other demand elasticities, price elasticity of supply; pp. 67–73 (applications) optional
- Project: your topic fixed by Thursday 15 October
- Problem set 1 (Blocks 1–2) handed out Thursday 15 October
- Practise in Sextant: the units the course page lists



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Slide 25 Before Thursday 15 October

No class on Monday 12 October; the next class is Thursday 15 October. Reading: Mankiw & Taylor chapter 3, pages 53 to 66, on the price elasticity of demand, the other demand elasticities and the price elasticity of supply; pages 67 to 73, the applications, are optional. Project: each group's question is fixed by Thursday 15 October, which is what today's meetings are for. Problem set 1, on Blocks 1 and 2, is handed out on Thursday 15 October. Practise in Sextant with the units the course page lists.

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